

ECO 4660-8666, International Economic Development, Fall 2024

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Problem Set 3

due Nov 25

1. Consider the following overlapping generations model where each generation lives two periods. Population size evolves as follow: $N'/N=(1+n)$ with $n=0.1$. The consumer will choose $c=c'$ whenever is possible. The income of a consumer when young is $y=150$. The consumer has no income when old: $y'=0$. The interest rate is $r=0$. There is a Pay-As-You-Go Social Security system funded by a tax t on the young. Retirement benefits are given out as a fixed amount b to each old consumer. There are no other taxes or government expenditures in this economy.
 - a. Find out the optimal consumption allocation in terms of t, b .
 - b. Suppose that the government levies a 20% social security tax on young, determine the social security benefit that would balance the social security budget. Note: the number of old people is different from the number of young people
 - c. Calculate lifetime wealth, optimal consumptions and savings under the social security system as in (b).
 - d. What would be lifetime wealth, consumptions and savings in this economy without social security?
 - e. EXPLAIN the impact of social security on welfare, consumption and savings in this model.
2. Use a diagram to explain
 - a. The impact of income growth on the household's choice of number of children.
 - b. The impact of closing gender wage gap on the household's choice of number of children.
3. Use words to explain the following concepts
 - a. In endogenous growth model (in the chapter discussing Education), we discuss that b, u matter to long-run growth. Please use concrete examples to explain how policies can affect above two parameters.
 - b. The limitation of Malthusian growth model;
 - a. Implicit and explicit cost of children;
 - c. Demographic transition;
 - d. Social and private rate of return of education.